Evidence-Separated Credit Scoring, Additive Explainability, and Shariah-Oriented Governance: A Research Prototype on a LendingClub Extract, with Independent UCI Methodology Context

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ABSTRACT This study presents a research prototype that separates credit-risk scoring, additive explainability, and Shariah-oriented governance evidence, and fuses them only at a frozen decision map. It is not Shariah certification, a fatwa, an automated compliance verdict, expert-validated Shariah decisioning, or a production credit engine, and it reports no fraud-detection performance. The primary dataset is a Zenodo LendingClub granting-model extract of 1,347,681 originated-and-resolved loans (US peer-to-peer, 2007–2018; not an Islamic-finance sample), with 941,267 training, 197,528 validation, and 208,886 test loans. Four LendingClub families—logistic regression, random forest, XGBoost, and an additive explainable boosting machine (EBM) with no interactions—were test-evaluated after a training-plus-validation refit (context M-4B); validation selected the EBM. Downstream probabilities come from a distinct training-only EBM with validation-only Platt scaling (context M-5). On the frozen test set, M-5 raw and Platt receiver-operating-characteristic area under the curve (ROC-AUC) both equal 0.6776693941139847; Platt Brier score is 0.15616920271522994 and expected calibration error (ECE) is 0.020267295489117824. The test set was not used to fit calibration. Explanations use an exact additive SHAP-style logit decomposition of the M-5 EBM; TreeSHAP was not used. The extract lacks contract-level Shariah fields. Dataset-level status is INSUFFICIENT_SCHEMA_FOR_APPLICANT_DETERMINATION; all 208,886 test rows receive INSUFFICIENT_INFORMATION—an evidence-schema limitation, not a compliance verdict. Frozen-score research maps assign 5,232 / 203,627 / 27, 0 / 208,859 / 27, and 61,855 / 0 / 147,031 approve / human-review / reject decisions under credit-only, fused, and single-threshold rules; predictive scores are identical across maps. Independent UCI logistic-regression and EBM results are methodology context only; UCI XGBoost is exploratory.

INDEX TERMS credit risk, evidence traceability, explainable boosting machine, probability calibration, Shariah-oriented governance.

I. INTRODUCTION

CONSUMER credit scoring is a statistical classification of applicants into higher- versus lower-risk classes [1]. Machine-learning classifiers have been benchmarked against traditional scorecards on institutional credit data [2], [3]. Separately, probability-of-default estimation has been argued to be more decision-relevant than a binary good/bad label in credit-card default research [4]. Peer-to-peer (P2P) lending extracts from the Lending Club platform have been used in prior risk and explainability studies [5], [6]; those studies do not evaluate the Zenodo granting-model extract executed here [7].

Two further literatures motivate the design of this prototype, without validating its test-set metrics. First, post-hoc probability calibration (sigmoid / Platt scaling and isotonic regression) is a standard method for mapping classifier scores to probabilities [8]–[10], and expected calibration error is a documented reliability summary [11]. Second, generalized additive models (GAMs) and glass-box boosting implementations provide additive structure that can be inspected feature-wise [12]–[14]. Additive SHAP attributions are defined for that class of explanations [15]; TreeSHAP is designed for tree ensembles rather than a first-order GAM [16].

A third motivation is governance and auditability rather than predictive gain. US adverse-action rules require specific principal reasons, including when complex algorithms are used [17]–[19]. Those instruments apply to US creditors; this study is a research prototype and is not presented as an Equal Credit Opportunity Act (ECOA) originator or as a compliance guarantee. In Islamic finance, official Shariah governance

frameworks for Islamic banking institutions in Pakistan and related documentary-evidence circulars describe board-level product approval and contract-level evidence requirements [20]–[22]. Separately, IFSB-10 records guiding principles for Shariah governance systems of institutions offering Islamic financial services [23]; it is cited as identity/FAQ context, not as a Pakistan Islamic-banking-institution circular and not as a ruling on this extract. Securities regulation in Pakistan further states that a security shall not be called Shariah compliant unless a Commission certificate is granted [24]. Those instruments do not license this software to certify loans in the executed LendingClub extract.

The executed design question is:

> Does separating credit-risk scoring, explainability, and Shariah-oriented governance into independently validated and auditable evidence layers, fused only at the final decision stage, provide a more transparent and defensible digital-financing pipeline than a monolithic end-to-end classifier without materially sacrificing predictive performance?

This paper treats that sentence as a **design question**, not as a proven operational-superiority result. On frozen calibrated probabilities, decision counts differ across maps while predictive scores remain identical. Governance is not claimed to improve ROC-AUC.

The evaluation examines two executed datasets only: the Zenodo LendingClub granting-model extract as the primary setting [7], and the UCI Default of Credit Card Clients table as an independent credit/calibration methodology context [4], [25]. German Credit and Australian Credit Approval are common literature benchmarks [26], [27] and were **not** executed. No fraud-detection experiment was executed; fraud detection is a distinct statistical task [28] and is not reported as a performance result of this study.

The remainder of this paper is organized as follows. Section II reviews related work. Section III states research-gap and contribution boundaries. Section IV describes methods. Section V records the experimental protocol and authorization classes. Section VI transcribes stored results. Sections VII–IX discuss, limit, and conclude.

II. RELATED WORK

Literature is used for methodological motivation and comparison with prior research. It is **not** used to claim that this project's test-set metrics have been independently validated. No 2023–2026 systematic review of machine-learning credit scoring was locked in the recorded literature sources; related work therefore uses the verified 1997–2022 scholarly sources and the recorded regulatory/standard instruments. No verified paper on real-time latency or throughput of loan engines was recorded; automated-lending discussion uses regulatory texts rather than systems-performance literature.

A. CREDIT SCORING AND MACHINE-LEARNING BENCHMARKS

Hand and Henley review statistical classification methods for consumer credit scoring and note that commercial scorecard detail is often unpublished [1]. Baesens *et al.* benchmark classification algorithms on real credit datasets [2]. Lessmann *et al.* update that benchmark with later algorithms and performance measures [3]. Those populations, periods, and pipelines differ from the executed LendingClub extract. They do not rank the models in this study.

Yeh and Lien compare data-mining techniques for probability of default on Taiwan credit-card clients and argue for reporting estimated default probabilities [4]. The corresponding UCI record is dataset 350 [25]. That table is the independent secondary dataset in this project. Yeh and Lien's ranking of methods on their metric is **their** result, not this project's UCI analysis.

Malekipirbazari and Aksakalli study random-forest risk assessment on Lending Club platform data [5]. Ariza-Garzón *et al.* study SHAP explanations of a P2P granting scoring model [6]. Those papers provide literature context for Lending Club / P2P modeling. They do not use the hashed Zenodo granting-model extract executed here [7], and they do not confirm this project's additive EBM attributions.

Common public credit benchmarks also include German Credit (UCI Statlog, $n = 1000$) [26] and Australian Credit Approval (UCI Statlog, $n = 690$) [27]. They appear here only as literature-context datasets. They were not acquired as executed study files and must not be read as additional empirical settings of this prototype.

B. EXPLAINABLE MODELS AND ADDITIVE ATTRIBUTIONS

Hastie and Tibshirani formulate GAMs as additive smooth functions of individual features [12]. Lou *et al.* describe GA2M models that may add pairwise interactions while remaining visualizable [13]. Nori *et al.* describe the Explainable Boosting Machine in InterpretML as a glass-box cyclic gradient-boosted GAM [14]. The present study froze EBM `interactions=0`; pairwise GA2M terms were not fitted to improve test-set scores. Citation of [13] is family background, not evidence that interactions were used.

Lundberg and Lee define SHAP additive feature attributions [15]. Lundberg *et al.* present TreeSHAP / TreeExplainer for trees, forests, and gradient-boosted trees [16]. TreeSHAP is not an appropriate explainer for a first-order EBM GAM; this project did not use it. Bussmann *et al.* apply Shapley-style explanations in an SME P2P credit-risk setting [29]. That population differs from the executed Zenodo LendingClub granting-model extract, and their pipeline is not this project's closed-form centered-term identity.

C. PROBABILITY CALIBRATION

Platt describes a sigmoid map from classifier scores to probabilities [8]. The recorded literature sources include **no DOI** for that chapter and do not verify an official publisher URL; identity is retained as recorded. Niculescu-Mizil and

Caruana compare Platt scaling and isotonic regression [9]. Zadrozny and Elkan describe isotonic-style calibration of ranking scores [10]. Guo *et al.* discuss calibration and ECE for modern neural networks on vision and document tasks [11]; that population is not credit scoring. These papers motivate calibration candidates. They do not supply the frozen Platt parameters a and b, and they do not validate the stored test-set Brier score or ECE.

D. BOOSTING ALGORITHMS

Chen and Guestrin describe XGBoost as a scalable tree-boosting system [30]. That citation is an algorithm description. It does not merge LendingClub XGBoost (context M-4B) with UCI XGBoost (context M-9.4-X), and it does not make the exploratory UCI XGBoost analysis confirmatory or overall superior.

E. FRAUD DETECTION (LITERATURE CONTEXT ONLY)

Bolton and Hand review statistical fraud detection as a task distinct from ordinary default prediction [28]. This project executed no fraud experiment. Literature [28] supports a **negative** claim: fraud-detection performance is not reported.

F. FAIRNESS, AUDITABILITY, AND AUTOMATED LENDING

Kozodoi, Jacob, and Lessmann study fairness criteria in credit scoring and document profit trade-offs [31]. This project did not execute fairness processors. Citation of [31] is related-work context only.

US Regulation B requires written adverse-action notices with specific principal reasons [17]. CFPB Circular 2022-03 states that complex algorithms do not excuse that requirement [18]. CFPB Circular 2023-03 states that sample forms are insufficient if they do not match factors actually used [19]. These texts motivate transparent, auditable scoring. They are not a finding that this prototype is production-ready or ECOA-compliant.

No verified real-time loan-engine latency paper is in the recorded literature sources. That is a literature gap, not a licence to invent systems-performance citations.

G. SHARIAH GOVERNANCE INSTRUMENTS (IDENTITY AND SCOPE)

Pakistan's Islamic Finance Policy Department issued a Shariah Governance Framework for Islamic banking institutions under IFPD Circular No. 08 of 2024 [20]. The SGF text effective 1 January 2025 describes prior Shariah Board approval of products and review of process flows [21]. The official SBP annex PDF was recorded as HTTP 403; a PICG-hosted public republication is used for the text [21]. SBP IBD Circular No. 01 of 2012 specifies invoice and alternative documentary evidence for Murabaha transactions of Islamic banking institutions [22]. That circular illustrates contract-level evidence that the LendingClub extract does not contain. It is not a rule for the executed LendingClub extract and is not an applicant-level ruling engine.

SECP Shariah Governance Regulations 2023 provide that no security shall be called Shariah compliant unless a Commission certificate is granted [24]. That instrument supports **prohibiting** certification language for this prototype. It excludes SBP banking companies and does not apply as a certificate authority for the executed LendingClub extract.

AAOIFI Shariah Standard (8) Murabahah and Standard (9) Ijarah exist as named standards [32], [33]. Operative clauses were not retrieved (password-protected / access-controlled viewer). Years for those identity pages are recorded as UNVERIFIED and are not filled here. AAOIFI also lists governance standards, including a 2024 Shariah Governance Framework listing [34]; clause text was not retrieved. IFSB-10 sets guiding principles for Shariah governance systems of institutions offering Islamic financial services [23]; a FAQ PDF was verified rather than a complete re-parse of the full standard body. None of these instruments is implemented as an automated fatwa engine, and none certifies this software.

III. RESEARCH GAP AND CONTRIBUTIONS

Prior work benchmarks classifiers [2], [3], studies P2P risk and SHAP granting models [5], [6], [29], and documents calibration and additive explanations [8]–[10], [12]–[16]. Official Shariah governance texts for Islamic banking institutions describe board-level product approval and contract-level documentation [20]–[22]. SECP securities regulation prohibits calling a security Shariah compliant without a Commission certificate [24]. IFSB-10 is cited as identity/FAQ context for IIFS governance principles [23], not as a Pakistan IBI circular. The recorded literature sources do **not** contain a verified end-to-end study that, on the hashed Zenodo LendingClub extract, (i) keeps credit scores, additive explanations, and Shariah-oriented evidence objects separate, (ii) fuses them only at a frozen decision map, and (iii) reports an independent UCI methodology context without treating UCI as architecture or Shariah validation.

That observation is a **gap relative to the recorded literature sources**, not a claim of absolute historical novelty. This paper does not call the architecture "novel."

The study develops a research prototype with the following contribution boundaries.

1) **Evidence-separated pipeline on LendingClub.** Credit-risk scoring, additive explanation, and governance evidence are separate objects. Fusion is a decision map on frozen calibrated probabilities `p_selected`, not a single end-to-end Shariah score.
2) **Chronological evaluation with frozen exclusions.** LendingClub uses a frozen issue-month split, train-only preprocessing for the downstream EBM, validation-only calibration selection, and one test-set report after freeze. No newly discovered leakage was recorded beyond the frozen exclusions. Hashes are recorded in split manifests. This is a documented split and feature-policy design, not a proof that every possible leakage path is absent.
3) **Transparent additive EBM reference.** The explainable reference is an EBM with `interactions=0` and an

exact additive logit-scale SHAP-style decomposition of **that** model. TreeSHAP was not used. Attributions are not causal and are not an approval criterion.

4) **Governance / evidence-gap prototype.** Implemented predicates record schema insufficiency on LendingClub. They do not determine applicant-level Shariah compliance. No rule is `EXPERT_VALIDATED`. AAOIFI operative clauses are unimplemented.
5) **Independent UCI methodology context.** Logistic regression and EBM are reported on UCI Default of Credit Card Clients as a separate credit/calibration methodology analysis. UCI XGBoost is an execution-time exploratory expansion (`DEVIATION-001`) and is not a primary validated model.

The following are **not** contributions of this paper: a certified Islamic-finance engine; a fraud detector; a new confirmatory test after the frozen decision maps; proof that governance improves discrimination; proof that XGBoost is superior overall; expert-validated Shariah decisioning; fairness guarantees.

Protocol hypotheses are design statements. They are not treated as proven results.

IV. METHODOLOGY

Four model contexts are kept separate throughout.

M-4B LendingClub XGBoost and M-9.4-X UCI XGBoost are distinct executions on different datasets and authorization classes. They are not one "XGBoost result."

A. RESEARCH ARCHITECTURE

The prototype has four documented layers (Fig. 1). These layers are transcribed from the executed governance and fusion records; they are not a reconstructed product architecture and do not claim a deployed production service.

1) **Credit-risk scoring.** A frozen probability of default. Downstream LendingClub decisions use M-5 `p_selected` (Platt-calibrated TRAIN-only EBM). They do not use the M-4B TRAIN+VALIDATION test EBM dump.
2) **Explainability.** Additive attributions of the M-5 EBM on the logit / `decision_function` scale. Platt is a later monotone map and is not inside the explanation.
3) **Shariah-oriented governance evidence.** Dataset-level (L1) and per-application (L2) predicates. Credit features are firewalled out of the Shariah evidence object. The loan `purpose` field is not treated as a contract type.
4) **Decision fusion.** A frozen map from credit band $\times$ L2 status to `{APPROVE, HUMAN_REVIEW, REJECT}` (G0; defined in Section IV-J). Fusion does not retrain models, refit Platt, or search thresholds.

The system is a research prototype for Shariah-oriented governance and evidence traceability. It is decision-support / evidence-gap handling. It does not replace a Shariah Board.

B. LENDINGCLUB DATASET

The primary executed dataset is the LendingClub granting-model extract `LC_loans_granting_model_dataset.csv`, Zenodo DOI `10.5281/zenodo.11295916`, version 0.1, license CC-BY-4.0 [7]. Official Zenodo creators are M. J. Ariza-Garzón, M. Sanz-Guerrero, and J. Arroyo Gallardo (Lending Club listed as data collector) [7]. File SHA-256 is `207d65b2712f775a91404496ab27c786ac0da6b0f977f849e6d5d8149e925acc`. The extract contains 1,347,681 rows $\times$ 15 columns.

The extract is depositor-filtered from a parent Kaggle Lending Club file. Kaggle is not a third executed dataset. The file contains originated-and-resolved loans; rejected applicants are absent (selection bias). The extract is recorded as US P2P 2007–2018. It is **not** Islamic-finance data and does **not** contain contract-level Shariah evidence fields sufficient for applicant-level Shariah compliance determination.

Target column: `Default`. Freeze coding: `1` = Charged Off, `0` = Fully Paid. A Zenodo target-documentation conflict was recorded; the freeze coding controls.

Time index used for splitting: `issue_month` derived from `issue_d`. Split hashes:

- TRAIN $n = 941267$, hash `02ba5877fdf189ba807a6b8e4463383fd04e86f6b3a65a1a5e9583f8b2d989d2`
- VALIDATION $n = 197528$, hash `d1585d22e101c717bf52426e8e625da960c975aa35db703a435bc4a7f8b3df91`
- TEST $n = 208886$, hash `4dfedb1ad7785fa3833047135b1883676f3e1b49aca9c87fcdb346e64fd489c6`

Prevalence differences by split (train 18.7%, validation 24.9%, test 21.2%) are descriptive context; M-5 TEST stored prevalence is 0.2117901630554465 with $n_{\text{positive}} = 44240$.

C. UCI CREDIT CARD DEFAULT DATASET

The independent secondary dataset is UCI Default of Credit Card Clients, UCI dataset 350, DOI `10.24432/C55S3H` [4], [25]. Zip SHA-256: `56c885f84457f6680f8438f02bfcdac9579323d8a94465ee5f26e32baa727602`. The table has 30,000 rows $\times$ 25 columns. Target: `default payment next month` (`1` = default next month). The UCI target is **not** equivalent to the LendingClub charged-off / fully-paid coding.

UCI has no calendar timestamps. Split: stratified 70/15/15 by target, seed 42:

- TRAIN $n = 20{,}999$, hash `70b1819de66fdae6793691e83224b5dc4e1956c021b582ffcf0a89791c9ac2f1`
- VALIDATION $n = 4{,}499$, hash `76a91b4b6fefbfd111d6a7c536ad222330c3bcf0f4358c259e40d4814382bcd4`
- TEST $n = 4{,}502$, hash `0783db42ac57a277f79379e0a62ef00f2a2b189f868ed993cebdf16def404126`

The tables were not merged. Undocumented UCI codes remain unlabeled. UCI analysis does not implement the Shariah governance layer and does not validate the LendingClub architecture. German Credit [26] and Australian Credit Approval

Model contexts kept separate throughout this study (presentation summary; no new metrics).				
Context ID	Name	Fit	Role	Used for LC decisions?
M-4B	LendingClub families (logistic regression, random forest, **LC XGBoost**, EBM)	After freeze, test metrics from TRAIN+VALIDATION refits	Family test after freeze (TRAIN+VALIDATION EBM refit)	**No**
M-5	TRAIN-only LendingClub EBM + validation-fit Platt	TRAIN-only EBM (`n_train=941267`); Platt on VALIDATION	Frozen LC decision pipeline	**Yes**
M-9.4-AE	UCI logistic regression and EBM	UCI-local splits	Independent methodology	**No**
M-9.4-X	UCI XGBoost	Execution-time expansion; `DEVIATION-001`	Exploratory UCI subsection only	**No**

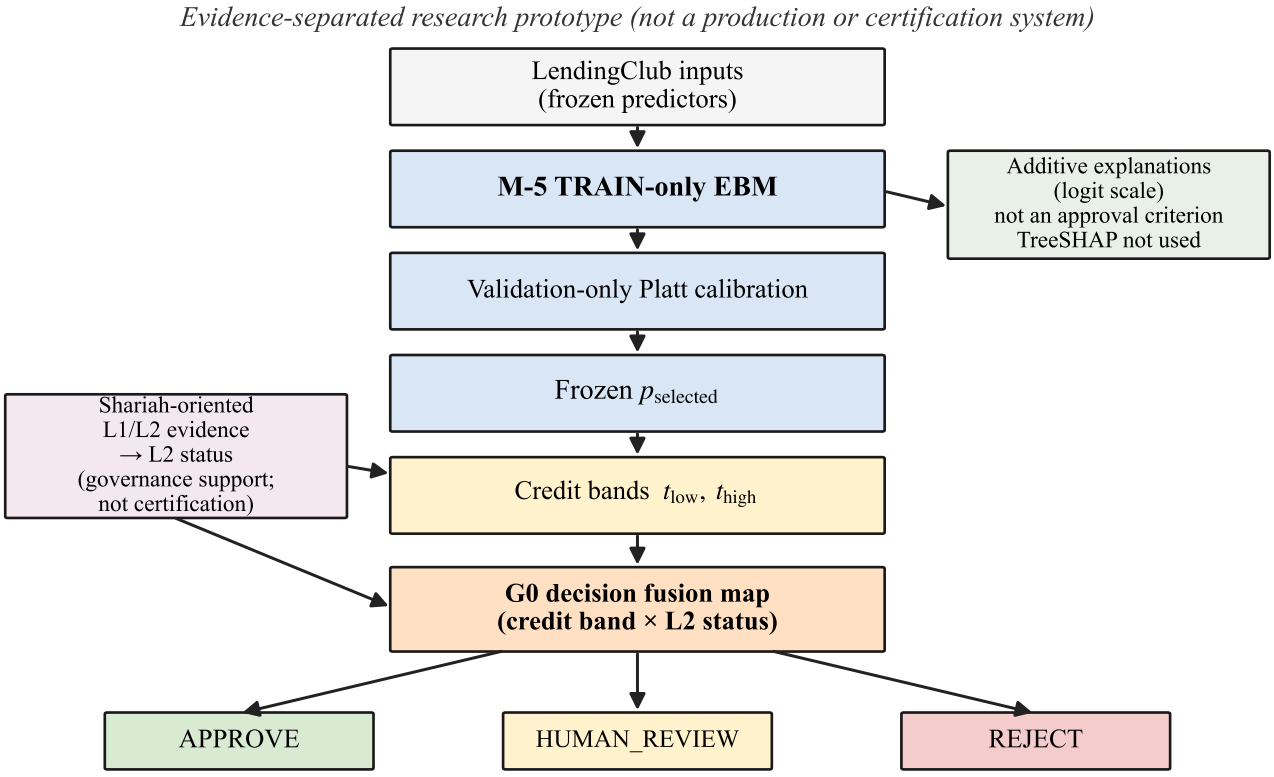


FIGURE 1. Evaluated evidence-separated research-prototype architecture. Center path: LendingClub inputs → M-5 TRAIN-only EBM → validation-only Platt → frozen $p_{selected}$ → credit bands → G0 fusion → `{APPROVE, HUMAN_REVIEW, REJECT}`. Right path: additive logit-scale explanations (not an approval criterion; TreeSHAP not used). Left path: Shariah-oriented L1/L2 evidence to L2 status (governance support, not certification). No fraud-detection layer was evaluated. This is not a production deployment.

[27] remain literature-context benchmarks only and were not executed.

D. FEATURE POLICY AND LEAKAGE CONTROLS

LendingClub primary predictors (both LC EBMs): `revenue`, `dti_n`, `loan_amnt`, `fico_n`, `emp_length`, `purpose`, `home_ownership_n`, `addr_state`, `zip_code`.

Excluded from the primary credit model: `id`, `issue_d`, `experience_c`, `title`, `desc`. Free-text `title` and `desc` were retained as a possible future branch; **no text model was built**. `addr_state` and `zip_code` are quasi-identifiers.

The LendingClub preprocessor for the downstream pipeline is fit on TRAIN only (`lendingclub_preprocessor.joblib`). High-cardinality `zip_code` is one-hot encoded for families that consume the expanded matrix (logistic regression, random forest, XGBoost). The EBM uses the nine imputed native columns rather than the one-hot expansion, because one-hot encoding would destroy additive per-original-feature structure. `dti_n==999` is unlabeled. Newly discovered leakage, beyond frozen exclusions, is recorded as none.

UCI retained columns are the standard LIMIT_BAL / demographics / PAY_ / *BILL_AMT* / PAY_AMT* block; `ID` is excluded.

Credit features are excluded from the Shariah evidence object (evidence firewall). `purpose` is not mapped to a Murabaha, Ijarah, or other contract type (purpose-invariance).

TABLE 1. Executed datasets (literature-only German Credit and Australian Credit Approval are excluded).

Field	LendingClub granting-model extract	UCI Default of Credit Card Clients
Role	**PRIMARY** credit-risk / decision-architecture setting	**Independent** credit-risk / calibration methodology only
Source	Zenodo DOI `10.5281/zenodo.11295916`; CC-BY-4.0; version 0.1 [7]	UCI dataset 350; DOI `10.24432/C55S3H` [4], [25]
Characterization	US P2P 2007–2018; not an Islamic-finance sample; originated-and-resolved loans, not the full application funnel	2005 Taiwan credit-card snapshot; no calendar timestamps
Full-file size	1,347,681 × 15	30,000 × 25
File / zip SHA-256	`207d65b2712f775a91404496ab27c786ac0da6b0f977f849e6d5d8149e925acc`	`56c885f84457f6680f8438f02bfcdac9579323d8a94465ee5f26e32baa727602`
Target	`Default` (1 = Charged Off, 0 = Fully Paid)	`default payment next month` (1 = default next month)
Split	Temporal by `issue_month`: TRAIN ≤ 2016-03; VAL 2016-04–2017-01; TEST ≥ 2017-02	Stratified 70/15/15 by target, seed 42
TRAIN / VAL / TEST *n*	941,267 / 197,528 / 208,886	20,999 / 4,499 / 4,502
TEST hash	`4dfedb1ad7785fa3833047135b1883676f3e1b49aca9c87fcdb346e64fd489c6`	`0783db42ac57a277f79379e0a62ef00f2a2b189f868ed993cebdf16def404126`
Merged?	Not merged with UCI	Not merged with LendingClub
Shariah contract-level fields	None sufficient for applicant-level determination	Governance layer not applied

E. TEMPORAL TRAIN/VALIDATION/TEST DESIGN

LendingClub is split **temporally** by `issue_month`:

- TRAIN: `issue_month <= 2016-03`
- VALIDATION: `2016-04` through `2017-01`
- TEST: `issue_month >= 2017-02`

Hash method: SHA-256 of sorted `issue_month` and `id` lines within each split.

UCI cannot use a calendar split. It uses a stratified random split with seed 42, documented as such. UCI therefore cannot support a temporal-generalization claim of the LendingClub type.

TEST is reporting-only after freeze for LendingClub family metrics (M-4B), calibration reporting (M-5), and decision counts on stored `p_selected`. Thresholds and calibrators were identified on validation. TEST was not used to choose the M-4B family, the Platt calibrator, or t_{low} / t_{high} / t^*.

F. BASELINE MODELS (M-4B)

M-4B compares four families on LendingClub. Hyperparameters were frozen before TEST. Validation seeds: 42, 123, 2024, 3407, 9999. Final TEST seed: 42.

Recorded configurations (seed-42 test `config` blocks):

- **Logistic regression:** `liblinear`, L2, $C = 1.0$, `class_weight='balanced'`, `max_iter=300`. StandardScaler on the numeric block, fit on the current training matrix.
- **Random forest:** 80 trees, `max_depth=12`, `min_samples_leaf=100`, `max_features='sqrt'`, `class_weight='balanced'`, sparse CSC input.
- **XGBoost (LendingClub only):** `binary:logistic`, 300 estimators, learning rate 0.05, `max_depth=4`, `min_child_weight=20`, subsample 0.8, `colsample_bytree=0.8`, `scale_pos_weight` computed on fitting labels (train+validation refit 4.061108666764441). Early stopping used validation during the train/val phase only; not used on TEST. This is **not** M-9.4-X.
- **EBM:** InterpretML EBM, `interactions=0`, `max_bins=256`, learning rate 0.02, `max_rounds=200`, `outer_bags=4`, nine native columns. No class-weight argument in EBM 0.7.8.

Primary validation criterion: mean PR-AUC across the five seeds. Best mean validation PR-AUC is 0.388473. Candidates within 0.005: EBM and XGBoost. Among those, lowest mean Brier then simplicity selected **EBM** (mean validation PR-AUC = 0.385841, Brier = 0.179907). TEST numbers were not used to choose the family. All completed families were still refit on TRAIN+VALIDATION and evaluated once on TEST after freeze.

The selected M-4B EBM dump path is `results/artifacts/final_explainable_boosting_machine_seed42.joblib`. Its SHA-256 is **not recorded**. That model is **not** used from the decision-fusion stage onward.

G. TRAIN-ONLY EBM AND CALIBRATION (M-5)

M-5 retrains an EBM on TRAIN only (`n_train=941267`) with frozen M-4B hyperparameters. Artifact: `results/calibration/artifacts/ebm_train_only_seed42.joblib`, SHA-256 `8d373df1b576ca74f498d2bfcc54653cbf43d37b0af69400a26e20ab98a3a0c4`. Preprocessor: `lendingclub_preprocessor.joblib` (train-only), not the M-4B train+validation preprocessor.

Protocol: validation-only calibration, no train+validation calibrator refit. Candidates stored: raw, Platt, isotonic. Calibrators are fit on VALIDATION raw probabilities and labels only. They are **not** fit on TEST and **not** refit on TRAIN+VALIDATION.

Platt scaling [8] maps raw scores by

$$p_{\text{platt}} = \sigma\big(a \cdot \text{logit}(p_{\text{raw}}) + b\big). \tag{1}$$

where σ is the logistic sigmoid, a and b are fit by unregularized logistic regression (`C=`∞, solver `lbfgs`) on $\text{logit}(p_{\text{raw}})$,

and frozen parameters are $a = 1.1988263209549705$, $b = 0.6697765046966161$, and `fit_split = validation`. Isotonic regression is stored (`y_min=0`, `y_max=1`, increasing, `out_of_bounds='clip'`) and is **not** selected [10].

Selection rule: validation Brier; 0.001 practical tie; simplicity order `raw > platt > isotonic`; declared before test. Best validation Brier is 0.17469431274375435 (isotonic). Platt validation Brier is 0.1748453678205707, within 0.001 of isotonic, so **Platt** is selected. All five validation seeds selected Platt.

TEST calibrated vector: `results/calibration/test/ebm_seed42_calibrated.csv`, SHA-256 `e15ddd2f85a6c3fc9d0a79eecb578f7a94ded8a7efc63d9d114212728b2469eb`. Column `p_selected` equals `p_platt` from (1). Decision fusion and later analyses consume this stored vector. ECE uses 10 equal-width bins on $[0, 1]$; empty bins are excluded from ECE and maximum calibration error (MCE).

Platt is a strictly increasing map of p_{raw}. Stored TEST ROC-AUC and PR-AUC of Platt therefore equal stored raw. Calibration is not claimed to improve discrimination.

Reliability-diagram image files named in the calibration report are **not retained** in the repository and are omitted from this manuscript. They are not regenerated from scores.

H. EXPLAINABILITY

The explainable reference is the additive M-5 EBM (`interactions=0`). Method:

- Exact additive interventional SHAP-style decomposition on the logit / `decision_function` scale:

$$\phi_i(x) = g_i(x_i) - \text{mean}_{\text{TRAIN}}\, g_i. \tag{2}$$

- TreeSHAP was **not** used; it is not mathematically appropriate for this first-order GAM [16].
- `shap.AdditiveExplainer` was not used (masker / mixed categoricals).
- Platt is not inside the explanation.
- Sample: 5,000 TEST rows, `RandomState(42)`, without replacement; labels not used to draw the sample. Not the full TEST set.

In (2), g_i is the fitted additive EBM term for feature i, and the mean is taken over TRAIN. The EBM is a transparent-model **reference**, not ground truth, not causal, and not an approval criterion [15]. Perfect rank agreement between SHAP and centered EBM terms is the identity of those centered terms, not independent confirmation from a second model class.

Documented provenance limitation (not silently rewritten): an explainability-report opening sentence refers to "the frozen Stage 4B model." Metadata (`model_id=ebm_train_only_seed42`; `stage5_model_path=results/calibration/artifacts/ebm_train_only_seed42.joblib`) identify the TRAIN-only M-5 EBM, not `results/artifacts/final_explainable_boosting_machine_seed42.joblib`. This paper follows the metadata.

I. SHARIAH-ORIENTED GOVERNANCE LAYER

Required description: research prototype for Shariah-oriented governance and evidence traceability. Not certification [24]. Not a fatwa. Not an automated Shariah-compliance verdict. Not expert-validated Shariah decisioning (`EXPERT_VALIDATED` = none).

Layer	Treatment in this study
A. Verified source / provenance	SBP SGF 2025 via IFPD Circular 08 of 2024 [20], [21]; SBP IBD Circular 01 of 2012 [22]; SECP SGR 2023 Regulation 7 [24]; AAOIFI SS 8 / SS 9 **identity** only [32], [33]; AAOIFI GS listing [34]; IFSB-10 identity/FAQ [23]
B. Implemented predicates	L1 dataset readiness vs L2 per-application evaluation; source-mapped SBP/SECP rules at `SOURCE_VERIFIED_PENDING_EXPERT_REVIEW`; AAOIFI identity/unavailability gates only; credit-feature firewall; purpose-invariance
C. Evidence / schema gaps	L1 `INSUFFICIENT_SCHEMA_FOR_APPLICANT_DETERMINATION`; L2 `INSUFFICIENT_INFORMATION` on all LC TEST rows; AAOIFI possession `NOT_VERIFIED`; one unsupported retired rule
D. Unvalidated / unavailable	AAOIFI operative clauses not retrieved or implemented; no `EXPERT_VALIDATED` rule

The fused `INSUFFICIENT_INFORMATION` result is an evidence/schema limitation. LendingClub does not contain invoices, possession records, or other contract-level Shariah fields contemplated by Murabaha documentary circulars [22]. `INSUFFICIENT_INFORMATION` is **not** a finding that an applicant is Shariah non-compliant, that a contract is compliant or non-compliant, or that the ML score determines Shariah compliance. The ML score does not determine Shariah compliance. Human review is not certification.

J. DECISION POLICIES

Thresholds were identified on validation under a researcher cost scenario, not operational cost:

- FN:FP $= 5 : 1$, $C_{\text{REVIEW}} = 0.25 \times C_{\text{FP}}$
- Two-threshold credit bands: $t_{\text{low}} = 0.06$, $t_{\text{high}} = 0.69$
- Monolithic single threshold: $t^* = 0.16$

Fig. 2 diagrams these frozen validation-derived thresholds. No new grid was searched at fusion time. Credit bands on stored p_{selected} follow

$$\text{band}(p) = \begin{cases} \text{LOW}, & p < t_{\text{low}}, \\ \text{HIGH}, & p \geq t_{\text{high}}, \\ \text{MEDIUM}, & \text{otherwise}. \end{cases} \tag{3}$$

with the frozen identification $t_{\text{low}} = 0.06$ and $t_{\text{high}} = 0.69$.

L1 schema status is dataset context and is **not** the G0 matrix input. L2 `overall_governance_status` is the matrix input.

Named maps (A3 identical to A4):

- **A0** `credit_only_selected_bands`: credit bands only.
- **A1** `fused_evidence_separated_v2`: G0 fusion.
- **A3/A4** `monolithic_credit_threshold`: single threshold t^* with no review band.

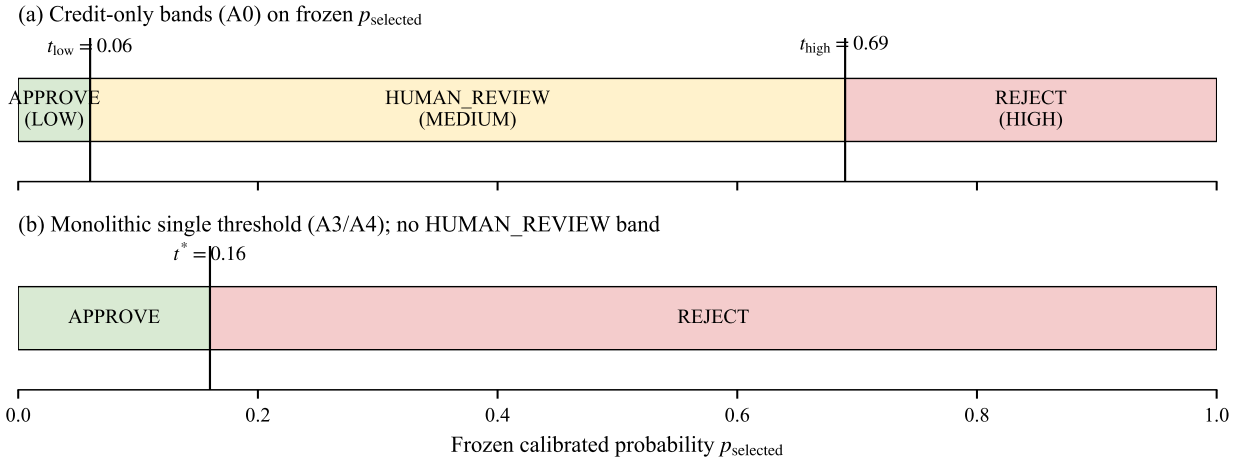


FIGURE 2. Frozen research decision thresholds on $p_{selected}$ (transcribed from `results/decision/v2/frozen_policy.json`). Panel (a): credit-only bands A0 with $t_{low} = 0.06$ and $t_{high} = 0.69$. Panel (b): monolithic A3/A4 with $t^* = 0.16$. `HUMAN_REVIEW` is a simulated research-policy band, not a collected human-review outcome and not a real loan-origination result. Fused map A1 applies the same credit bands through the G0 matrix (Table 9).

TABLE 9. G0 fusion matrix. `INSUFFICIENT_INFORMATION` is not converted to `PASS` or to religious non-compliance. The same matrix is not repeated as a separate figure.

Governance (L2)	LOW	MEDIUM	HIGH
`NOT_STRUCTURABLE_WITHOUT_REDESIGN`	REJECT	REJECT	REJECT
`INSUFFICIENT_INFORMATION`	HUMAN_REVIEW	HUMAN_REVIEW	REJECT
`REVIEW`	HUMAN_REVIEW	HUMAN_REVIEW	REJECT
`PASS`	APPROVE	HUMAN_REVIEW	REJECT

These are policy analyses on frozen `p_selected`, not real-world approval performance. Maps are not ranked as best / optimal / winner.

Recorded freeze SHA-256 values: v2 `b874782326a8f20ed9bb7df1aa5ed1c75f883d8b6cba76b68991181d50b1604b`; v1 `4d3da6a9e5bac2b38b546c7a5a60e870bf83f30adcadf67f067f99951e32f0c4`; decision-count table `90759eb72f1173b40b5b9363c21b0ef3d6f1cde5042748eb57871d49fb73d5ee`.

V. EXPERIMENTAL PROTOCOL

The empirical program is sequential and frozen. Later analyses do not retune thresholds, refit calibration, or reopen family selection.

Stage	What was frozen	Test-set role	Authorization class for later citation
Data / split	Acquisition, hashes, feature policy, splits	Split construction	Data provenance
Preprocessing	Train-only LC preprocessor	Not a predictor test	Methodology
M-4B	Four-family LC models; EBM selected on validation	One family test after TRAIN+VAL refit	Family test after freeze
M-5	TRAIN-only EBM; validation-only calibrator selection	Reporting of raw/Platt/isotonic	M-5 pipeline reporting
Explainability	Attributions of M-5 EBM	Descriptive sample $n = 5000$	Descriptive of that model
Governance	L1/L2 predicates	Governance evidence, not certification	Prototype governance
Thresholds	Identification on validation	None for new search	Freeze-backed identification
Fusion	G0 fusion on stored `p_selected`	Decision counts	Freeze-backed policy counts
Map ablation	Ablation of frozen maps	Previously exposed test set	Class B documentation-gap authorization
Cost sensitivity	Assumed-r cost sensitivity	Stored counts under alternative r	Class C exploratory
Hypothetical G1/G3	Hypothetical allocation	Hypothetical allocation	Class C exploratory; not Shariah findings
UCI LR+EBM	UCI-local family test	UCI test	Approved independent methodology (**M-9.4-AE**)
UCI XGBoost	Execution-time UCI family expansion	UCI test	Class C `DEVIATION-001`; **M-9.4-X** exploratory
Integrity check	Hash / identity check	No new scores	Integrity; not confirmatory science
Year mix	2017/2018 mix of frozen maps	Post-hoc slices	Exploratory / descriptive post-hoc
Evidence-gap algebra	Algebra of A1 review	Existing L2 statuses	Analysis-only; not Shariah validation
Side-by-side citation	Citation comparison LC vs UCI	No new AUC	Analysis-only

M-4B and M-5 test confidence intervals are percentile bootstrap, 1000 row resamples, seed 42, not used for tuning. UCI XGBoost remains exploratory under `DEVIATION-001` and is not a primary validated model. Metric wrappers used in the stored artifacts are ROC-AUC = `sklearn.metrics.roc_auc_score`, with PR-AUC, Brier, and log loss as recorded in the project metric-definition notes.

VI. RESULTS

Every numeral below is transcribed from a stored artifact. No new test metric is computed in this manuscript.

A. LENDINGCLUB BASELINE RESULTS

Context: **M-4B**. TEST $n = 208{,}886$. TRAIN+VALIDATION refit after family freeze. These metrics are **not** the downstream decision scores.

The gap is not a calibration effect on rank: M-5 Platt TEST ROC-AUC equals stored raw. It is not an unresolved error.

Percentile 95% bootstrap CIs are stored with the same metrics (1000 resamples). For the selected EBM: ROC-AUC CI $[0.6772520975812178, 0.6828319764466856]$; PR-AUC CI $[0.3492795049814313, 0.3577134296795478]$; Brier CI $[0.15586719460814155, 0.1579539606709722]$.

ECE is **not stored** for Stage 4B logistic regression, nor for the other Stage 4B families (random forest, XGBoost, EBM), in the Stage 4B test-metric artifact. No Stage 4B ECE value is calculated or inferred here.

TEST PR-AUC is lower than validation PR-AUC for every family. Descriptive year slices of the M-4B TEST EBM (not used to retune): 2017 ROC-AUC 0.677804, PR-AUC 0.377247, Brier 0.168236; 2018 ROC-AUC 0.675593, PR-AUC 0.276294, Brier 0.125866.

M-4B LC XGBoost TEST ROC-AUC 0.6854404067747387 is a Stage 4B family result. It is not the decision-fusion model and is not M-9.4-X.

Fig. 3 plots the stored TEST ROC-AUC values for the four M-4B families and the distinct M-5 TRAIN-only EBM (raw and Platt ROC-AUC are identical). The figure does not collapse the two EBM fits.

B. CALIBRATION RESULTS

Context: **M-5**. TEST $n = 208{,}886$; $n_{\text{positive}} = 44240$; prevalence 0.2117901630554465.

Platt TEST ROC-AUC equals raw. Isotonic is piecewise constant and slightly changes TEST ranking (PR-AUC $0.35096314719303284 \rightarrow 0.3462755761696744$).

Paired TEST differences (calibrated - raw), not recomputed: Platt Brier -0.001767 (bootstrap CI $[-0.002005, -0.001508]$); Platt ECE -0.017267 (CI $[-0.020730, -0.014170]$). Intervals are not interpreted as hypothesis-test p-values in the source record.

Descriptive temporal slices under frozen Platt (not a new confirmatory test): 2017 $n = 152726$, Platt ECE 0.005096425775164428; 2018 $n = 56160$, Platt ECE 0.06152417312325683. On 2018, Platt increases Brier, log loss, and ECE relative to raw. The validation window is 2016-04 through 2017-01; this is classified as temporal transfer failure, not a reason to recode the frozen calibrator.

Reliability diagrams named in the calibration report (10 equal-width ECE bins) are historical path references. Those image files are not retained and are not regenerated from scores. Fig. 4 instead plots the **stored** TEST risk-range table (`results/calibration/tables/stage5_risk_ranges.csv`): mean predicted probability versus observed default rate for raw and selected Platt. That table uses the five stored ranges 0.00–0.10, 0.10–0.20, 0.20–0.30, 0.30–0.50, and 0.50–1.00; it is not a reconstruction of the omitted 10-bin ECE reliability image.

C. EXPLAINABILITY RESULTS

Context: M-5 EBM; descriptive attributions; sample $n = 5000$.

Agreement: `overlap_top3=1.0`, `overlap_top5=1.0`, `overlap_top9=1.0`, Spearman $= 1.0$, Kendall $= 1.0$, `additive_reconstruction_max_abs_error` $= 8.881784197001252 \times 10^{-16}$. This identity is expected for centered additive terms. It is not independent confirmation from a second model class and is not a causal ranking of default drivers. Fig. 5 visualizes the same stored mean-absolute series.

Local case tables exist as stored artifacts; they are not reproduced in full here. They remain attributions of this EBM, not approval criteria.

D. DECISION-POLICY RESULTS

TEST $n = 208{,}886$; $n_{\text{positive}} = 44240$.

Predictive scores are the same stored `p_selected` across maps. Governance is not claimed to change ROC-AUC. Maps are not ranked as winner.

The class-B map ablation restates that decision counts differ while scores are identical. It is a descriptive re-analysis of previously exposed TEST, not a new confirmatory experiment. Class-C cost sensitivity and hypothetical G1/G3 allocations are not reproduced as main-text rankings. G1/G3 are hypothetical engineering maps, not empirical Shariah findings.

E. UCI INDEPENDENT ANALYSIS

Context **M-9.4-AE** (approved written scope) and **M-9.4-X** (exploratory). UCI TEST $n = 4502$; $n_{\text{positive}} = 996$; prevalence 0.22123500666370502. This subsection does **not** validate the LendingClub architecture or Shariah governance.

Logistic-regression and EBM family-raw ECE are **not stored**. Those ECE values are not calculated or taken from another family's file.

The XGBoost ECE cell is transcribed from the selected-family calibration table, row `method=raw`, `split=test`, `seed=42`: ECE 0.20065704943022225. The selected family is XGBoost. The stored ROC-AUC, Brier, and log loss in that raw row match the family-raw XGBoost row. This ECE belongs only to context **M-9.4-X** (`DEVIATION-001`, exploratory). It is not LR/EBM ECE, not a primary validated result, not a ranking or superiority claim, and is

TABLE 2. Two LendingClub EBMs (must not be collapsed).

	M-4B TEST EBM	M-5 raw / Platt TEST
Training split	TRAIN+VALIDATION refit	TRAIN only (`n_train=941267`)
Preprocessor	`preprocessor_train_plus_validation.joblib`	`lendingclub_preprocessor.joblib`
Model dump	`results/artifacts/final_explainable_boosting_machine_seed42.joblib` (SHA-256 **not recorded**)	`results/calibration/artifacts/ebm_train_only_seed42.joblib` (SHA-256 `8d373df1b576ca74f498d2bfcc54653cbf43d37b0af69400a26e20ab98a3a0c4`)
TEST n	208,886	208,886 (same chronological test split)
ROC-AUC	**0.6799408767429772**	**0.6776693941139847**
Gap classification	**EXPLAINED** (different fitted models)	Same
Used from decision fusion onward	**No**	**Yes** (`p_selected`)

TABLE 3. M-4B LendingClub TEST metrics (seed 42; TRAIN+VALIDATION refit).

Family	ROC-AUC	PR-AUC	Brier	Log loss
`logistic_regression`	0.6763559009686606	0.3406477564928828	0.21998431914321218	0.6354126164492524
`random_forest`	0.6609260816246479	0.33258566142201473	0.23597628092130948	0.6648999360582141
`xgboost` (LC only)	0.6854404067747387	0.3576696570242051	0.2136380371614269	0.6149464126269267
`explainable_boosting_machine`	0.6799408767429772	0.3534066581608194	0.15684495750106664	0.48493888407781477

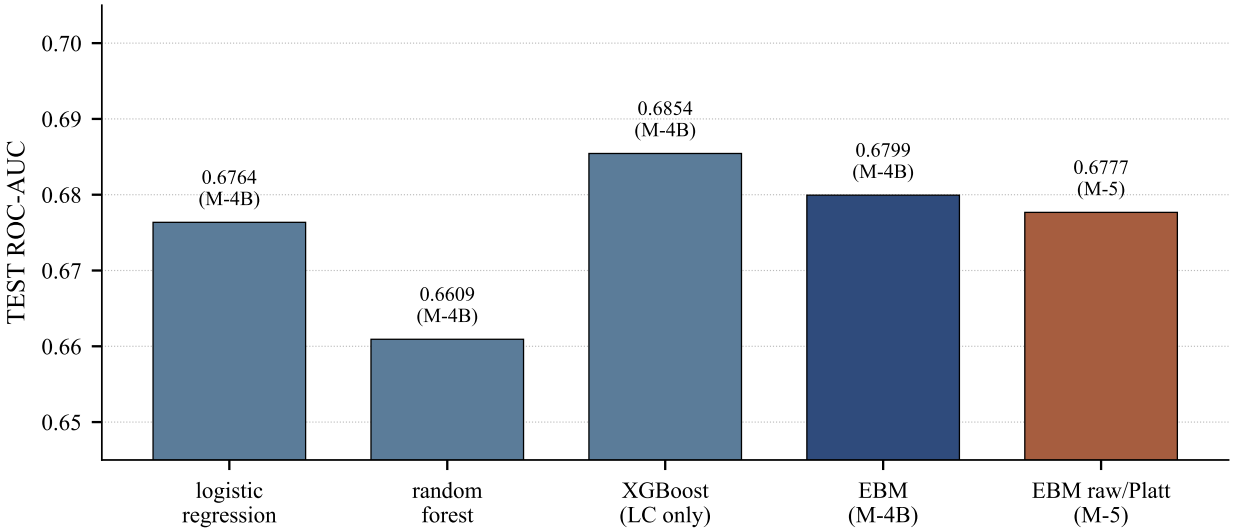


FIGURE 3. Stored LendingClub TEST ROC-AUC values. The first four bars are context M-4B (TRAIN+VALIDATION refit family TEST; seed 42). The fifth bar is context M-5 (TRAIN-only EBM; Platt and raw ROC-AUC both equal 0.6776693941139847). Values are transcribed from `results/metrics/stage4b_test_metrics.json` and `results/calibration/metrics/stage5_test_metrics.json`. M-4B and M-5 are different fitted models and must not be averaged.

TABLE 4. M-5 calibration metrics (TRAIN-only EBM; calibrators fit on validation only).

(a) Validation ($n = 197{,}528$; $n_{\text{positive}} = 49163$; prevalence 0.24889129642379815).

Method	Brier	Log loss	ECE	MCE	ROC-AUC	PR-AUC
raw	0.1799084821590894 7	0.5412792594262 45	0.0628621353713 6044	0.6023420058578064	0.6656900641018727	0.38583491323890046
Platt (selected)	0.1748453678205707	0.5278455947278384	0.0022124994740890247	0.0264319203524 6955	0.6656900641018727	0.38583491323890046
isotonic (not selected)	0.17469431274375435	0.5273917354891987	$1.5969769753490 43 \times 10^{-17}$	$1.1102230246251565 \times 10^{-16}$	0.6661439550959414	0.3823621419016197

(b) TEST (reporting only).

Method	Brier	Log loss	ECE	MCE	ROC-AUC	PR-AUC
raw	0.1579357158246714	0.4880855738039092	0.03753450535939786	0.38427225714072644	0.6776693941139847	0.35096314719303284
Platt (selected)	0.15616920271522994	0.48357015583149326	0.020267295489117824	0.1869468566677409	0.6776693941139847	0.35096314719303284
isotonic (not selected)	0.1561969364282068	0.48360807473639705	0.020046149884935992	0.30756843800322053	0.6775144297473418	0.3462755761696744

not used to alter LendingClub conclusions. No file exists at `results/calibration/uci_test_calibra`

TABLE 5. **Global mean absolute attribution (logit scale; sample of 5,000 TEST rows).**

Feature	shap_mean_abs	shap_rank	ebm_mean_abs	ebm_rank
fico_n	0.3072104502059198	1	0.3072104502059198	1
loan_amnt	0.2887712351465414	2	0.2887712351465414	2
dti_n	0.15084749758493932	3	0.15084749758493932	3
revenue	0.1449737450602624	4	0.14497374506026237	4
home_ownership_n	0.12252891641707762	5	0.12252891641707761	5
zip_code	0.11979470116252551	6	0.11979470116252551	6
purpose	0.10257411348554467	7	0.10257411348554467	7
emp_length	0.03984046466379107	8	0.03984046466379107	8
addr_state	0.007289702255172925	9	0.007289702255172925	9

TABLE 6. **TEST decision counts on frozen** `p_selected`**.**

Map	Artifact architecture	APPROVE	HUMAN_REVIEW	REJECT	Approval rate
A0	credit_only_selected_bands	5232	203627	27	0.025047154907461486
A1	fused_evidence_separated_v2	0	208859	27	0.0
A3/A4	monolithic_credit_threshold	61855	0	147031	0.2961184569573835

TABLE 7. **UCI TEST family-raw metrics and exploratory selected pipeline. Family-raw ROC-AUC, PR-AUC, Brier, and log loss are transcribed from the family-raw table, which has no ECE column.**

Family	Context	ROC-AUC	PR-AUC	Brier	Log loss	ECE
logistic_regression	M-9.4-AE	0.7572778277972128	0.5255105923338794	0.18709593385802795	0.5707348100103825	not stored
explainable_boosting_machine	M-9.4-AE	0.7688563724378406	0.53737573942064	0.13743237899672608	0.4384226822615689	not stored
xgboost	M-9.4-X exploratory	0.7733509623204742	0.5456637462205797	0.18173602230220492	0.5518604686417851	0.20065704943022225 (selected-family raw; note below)

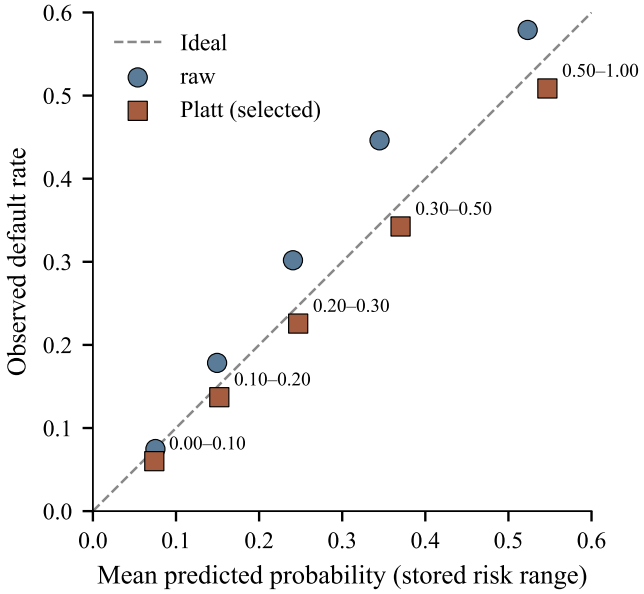


FIGURE 4. **Stored TEST risk-range calibration for the M-5 EBM: mean predicted probability versus observed default rate for raw and Platt (selected), transcribed from** `stage5_risk_ranges.csv`**. Dashed line: ideal equality. Labels mark the Platt range intervals. This is not the omitted 10-bin ECE reliability diagram and does not recompute ECE.**

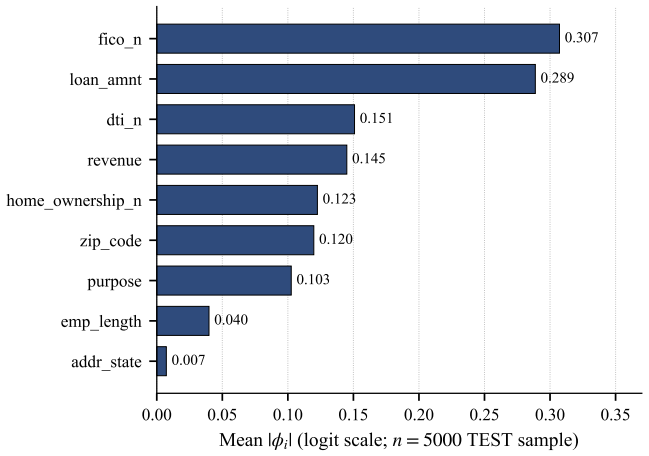


FIGURE 5. **Global mean absolute additive attributions on the logit scale for the M-5 EBM (Eq. (2)), transcribed from** `results/explainability/tables/table1_global_attribution.csv` **(sample of 5,000 TEST rows). Values equal the centered EBM term magnitudes under Eq. (2). TreeSHAP was not used. Attributions are not causal and are not an approval criterion.**

tion.csv. The stored side-by-side citation comparison records xgboost_ratified=false and keeps XGBoost off the primary comparison table.

Exploratory selected UCI pipeline: **XGBoost + isotonic**, class EXECUTION-TIME EXPANSION, EXPLORATORY (DEVIATION-001). TEST ROC-AUC 0.7708780071798891; PR-AUC 0.5224208622925242; Brier 0.1368528550434721; log loss 0.449533259034068; ECE 0.01744321566180452. The declared PR-AUC-first rule selected XGBoost **within that exploratory UCI analysis**. That selection does **not**

make XGBoost the primary validated model, the confirmed best model, an overall superior model, or a pre-registered confirmatory outcome. The five-seed XGBoost validation table is degenerate. M-9.4-X is not merged with M-4B LC XGBoost [30].

Returning to the LendingClub test set, Table 8 reports credit-band default rates on the same frozen scores used in Table 6.

HIGH band $n = 27$ is small; the interval is wide. Fused REJECT coincides with HIGH under observed L2 statuses ($n = 27$, default rate 0.5185185185185185). Fused HUMAN_REVIEW $n = 208859$, default rate 0.21175051111036633. Every populated governance-credit cell is `INSUFFICIENT_INFORMATION` × LOW: 5232 HUMAN_REVIEW; MEDIUM: 203627 HUMAN_REVIEW; HIGH: 27 REJECT. A1 HUMAN_REVIEW = 208,859 is LOW+MEDIUM × all-TEST insufficiency under G0, **not** a Shariah score.

F. DESCRIPTIVE / POST-HOC ANALYSES

These activities are **not** confirmatory test experiments.

The integrity check is a provenance review. A0/A1/A3 share stored `p_selected`. Freeze hashes match recorded values. That check does not generate an independent scientific result.

Year mix of the frozen TEST vector is exploratory / descriptive post-hoc (non-confirmatory): 2017 $n = 152726$ (default rate 0.2317483598077603); 2018 $n = 56160$ (default rate 0.15751424501424502).

This table was not used to year-tune policy. HIGH-band small counts remain a caveat (overall HIGH $n = 27$).

A1 review = 208,859 = 5232 + 203627. The algebra decomposes existing governance evidence. It does **not** validate Shariah compliance and does not mean applicants are religiously non-compliant.

A stored side-by-side citation comparison classifies the M-4B vs M-5 EBM ROC-AUC gap as **EXPLAINED**. It is not a new AUC test and not a model ranking.

VII. DISCUSSION

The stored artifacts show that an additive TRAIN-only EBM can be calibrated on validation, frozen, and used as a single probability stream for several decision maps without refitting on TEST. Discrimination of that downstream pipeline on LendingClub TEST is ROC-AUC 0.6776693941139847. M-4B family TEST after a TRAIN+VALIDATION refit is numerically nearby but is a different fit (EBM ROC-AUC 0.6799408767429772; LC XGBoost 0.6854404067747387) and is not a licence to swap joblibs into the decision-fusion pipeline.

Stored TEST Brier and ECE are lower under the frozen Platt map than under raw (paired differences). Those stored differences are a probability-mapping outcome [8], [9]; they are not a hypothesis-test claim. Stored TEST ROC-AUC did not change under Platt. 2018 ECE deterioration under frozen Platt is a descriptive temporal limitation, consistent with a validation window that ends in January 2017.

Explainability results are internally consistent with an additive GAM [12]– [15]. On the 5,000-row sample, the highest mean-absolute attributions are `fico_n` and `loan_amnt`. Perfect SHAP–EBM agreement should not be advertised as a second-model confirmation.

The governance layer refused to convert missing contract evidence into `PASS` or into a religious non-compliance label [22], [24]. Consequently A1 assigns APPROVE to nobody on this extract. That is an evidence-gap outcome, not a finding that LendingClub borrowers are Shariah non-compliant. A0 still yields only 5,232 TEST APPROVE assignments under $t_{\text{low}} = 0.06$, fewer than the 61,855 APPROVE assignments under the monolithic $t^* = 0.16$ map. Those differences are decision-map differences, not score differences.

UCI LR/EBM TEST ROC-AUC values in Table 7 (0.7572778277972128 and 0.7688563724378406) are not comparable as a claim that the LendingClub architecture generalizes. Targets, populations, split types, and feature sets differ [4], [25]. Exploratory UCI XGBoost must not be used to declare overall superiority [30].

Fairness processors were not run [31]. Fraud detection was not studied [28]. Production, real-time, or ECOA compliance is not evidenced [17]–[19]. Expert Shariah validation is absent.

The design question in Section I is therefore only partially addressable with stored evidence: separation of objects is recorded in the stored layer artifacts; ‘‘without materially sacrificing predictive performance’’ cannot be read as a governance-induced AUC change, because scores are identical across A0/A1/A3; operational defensibility in Islamic-finance institutions is not shown, because contract-level fields and expert review are absent.

VIII. LIMITATIONS

The following limitations are recorded in the executed evidence. None of the scientifically necessary limitations is silently dropped.

1) No `.git` repository / `git_commit` is `null` throughout recorded manifests.
2) SHA-256 of the M-4B final EBM joblib is not recorded.
3) Fusion audit records lack `row_ref`; identity in the integrity check is positional.
4) Stored pytest summaries do not independently lock every historical TEST point estimate.
5) The UCI XGBoost five-seed validation table is degenerate.
6) M-9.4-X remains `DEVIATION-001` (execution-time expansion, exploratory); no successor amendment ratifies it as confirmatory.
7) Map ablation authorization is class B (documentation gap), not a new confirmatory test.
8) Cost-sensitivity and hypothetical G1/G3 analyses are class C exploratory; G1/G3 are not empirical Shariah findings.
9) AAOIFI operative clauses are unavailable and unimplemented [32], [33].
10) No rule is `EXPERT_VALIDATED`.

TABLE 8. Credit-band default rates on LendingClub TEST.

Band	n	n_{default}	Default rate	Wilson 95% CI
LOW	5232	196	0.03746177370030581	[0.0326463743097964 7, 0.04295591281821748]
MEDIUM	203627	44030	0.21622869265863565	[0.21444596353189307, 0.21802212876714 69]
HIGH	27	14	0.5185185185185185	[0.3398532458621043, 0.6925704935029092]

TABLE 10. Year mix of frozen maps (post-hoc; not a confirmatory temporal test).

Year	Map	n	APPROVE	HUMAN_REVIEW	REJECT
2017	A0	152726	3339	149366	21
2017	A1	152726	0	152705	21
2017	A3/A4	152726	42443	0	110283
2018	A0	56160	1893	54261	6
2018	A1	56160	0	56154	6
2018	A3/A4	56160	19412	0	36748

TABLE 11. Observed L2 × credit band on TEST (descriptive; not a Shariah ruling).

Credit band	L2 status	n	G0 decision
LOW	`INSUFFICIENT_INFORMATION`	5232	HUMAN_REVIEW
MEDIUM	`INSUFFICIENT_INFORMATION`	203627	HUMAN_REVIEW
HIGH	`INSUFFICIENT_INFORMATION`	27	REJECT
All other L2 statuses	—	0	(unpopulated)

11) TRAIN+VALIDATION concatenated n for M-4B is not stored as a single field.
12) A Zenodo target-documentation conflict was frozen in coding.
13) LendingClub originated-and-resolved selection bias; no reject inference.
14) UCI and LendingClub targets are not equivalent; UCI undocumented codes unlabeled.
15) 2018 Platt ECE deterioration; no nested TRAIN calibrator.
16) Explanation sample $n = 5{,}000$, not full TEST.
17) HIGH credit band $n = 27$.
18) ECE is not stored for any M-4B family. UCI family-raw ECE is not stored for logistic regression or EBM. An exploratory selected-family **raw** ECE is stored separately for UCI XGBoost and is reported only under M-9.4-X / `DEVIATION-001`.
19) High-cardinality zip one-hot; `dti_n==999` unlabeled.
20) Explainability-report opening language versus TRAIN-only metadata: documentation/provenance limitation; historical report files were not rewritten.
21) LendingClub lacks contract-level Shariah fields; the evidence-gap algebra does not validate compliance; `INSUFFICIENT_INFORMATION` is not a Shariah ruling.
22) Literature sources: Platt 1999 has no verified DOI/URL in the literature package [8]; Zenodo creators are recorded from the official Zenodo record [7]; no 2023–2026 systematic ML-credit review; no real-time latency paper; IFSB-10 full body not re-parsed [23]; AAOIFI SS 8/9 years UNVERIFIED [32], [33].
23) Fairness not empirically evaluated [31].
24) No fraud experiment [28].
25) Thresholds $0.06/0.69/0.16$ are researcher-scenario identifications, not official operational costs.
26) Calibration reliability-diagram image files are not retained in the repository and are omitted from this manuscript (not regenerated).
27) German Credit and Australian Credit Approval are literature/context datasets only [26], [27].
28) Year-mix and evidence-gap analyses are descriptive / non-confirmatory.

IX. CONCLUSION

This study presents a research prototype that separates credit-risk scoring, additive explainability, and Shariah-oriented governance evidence, and fuses them only at a frozen decision map. On a hashed chronological LendingClub extract, a TRAIN-only EBM with validation-only Platt calibration yields frozen TEST ROC-AUC 0.6776693941139847. Research policy maps applied to the same `p_selected` change APPROVE / HUMAN_REVIEW / REJECT counts (A0: 5232 / 203627 / 27; A1: 0 / 208859 / 27; A3/A4: 61855 / 0 / 147031) without changing predictive scores. Those counts are not production origination outcomes. A1's all-review-but-27-reject pattern follows from `INSUFFICIENT_INFORMATION` on every TEST row—an evidence/schema limitation of the executed LendingClub extract that lacks contract-level Shariah fields—not from a finding of applicant non-compliance.

An independent UCI logistic-regression / EBM analysis provides methodology context only. Exploratory UCI XGBoost is not a primary validated model. The prototype is not Shariah certification, a fatwa, an automated compliance verdict, or a substitute for Shariah experts. It is not a production credit

engine and reports no fraud-detection performance.

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